



Apple to detect mental health

iPhones may soon detect depression, autism, and cognitive decline.
<https://digit.in/oct21-05>



Google buys new offices!

Google buying new office building for \$2.1 billion in New York
<https://digit.in/oct21-06>

Riemann Hypothesis

One of the most famous 'unsolved' mathematical questions.

Jaykrishna Gadhavi
feedback@digit.in

M

Mathematicians consider the solution to the Riemann

Hypothesis as the

Holy Grail of mathematics. Clay university has announced a price of one million dollars to whoever solves this question. Bernhard Riemann was a German mathematician who wrote papers on analysis and differential geometry, with his work on the latter forming the mathematical basis for Einstein's theory of relativity. Riemann wrote only one paper on number theory and it became the Holy Grail for mathematicians. What does it say? What is it about? Let's dive deep into it. There is a million dollar bounty on this equation getting solved, but the solution here is actually more valuable to mathematicians.

Riemann's hypothesis is all about prime numbers, distribution of primes, functions, and complex numbers. As we all know, prime numbers are the numbers that are only divisible by 1 and themselves.

Prime numbers seem to randomly pop up on the number line. There are two facts about the distribution of prime numbers. The first is that they are the most arbitrary and ornery objects studied by mathematicians: they grow like weeds among the natural numbers, seeming to obey no other law than chance, and nobody can predict where the next one will sprout. The second fact is even more astonishing, for it states just the opposite: that the prime numbers exhibit stunning regularity, that there are

laws governing their behavior, and that they obey these laws with almost military precision.

To introduce complex numbers, I'll say that mathematicians were tired of using real numbers which they found on the number line and they wondered "What is the square root of -1?". So, they thought of a very basic number that cannot be represented on the number line but only existed in their heads, $\sqrt{-1}$, which is represented as " i ", and called it an "Imaginary number". The real numbers, -3, -2, -1, 0, 1, 2, 3,....

